

DAY - 01.

→ Python → Interpreter language

↳ Executes the code line by line.

→ Default Input value = String. (slow).

Data Types:

- Int
- Float
- Boolean
- String

seq. dt.

Sets

Tuples

Dictionary

List

→ sequential data type
Range

Operators:

Bitwise:

Left Shift

Membership op.
↳ in
↳ not in

For Example: $n = 6$, $k = 3$

$2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

1 1 0

1 1 0 0 → $k=1$

1 1 0 0 0 → $k=2$

1 1 0 0 0 0 → 48 → $k=3$

Formula: $n \cdot 2^k$

Right Shift

$$\frac{n}{2^k}$$

Conditional Statements :

if
elif
else

H.W :

Does Python support "default"

Control Structures :

break , continue

Loops :

for loop : membership op.
for i in range(s, e, skip)

Output Formatting :

By default Python prints each output in a new line

To control we use `end = " "`

Formatted String
`print(f" ")`

Program to print if a no. is Prime or not.

for i in range(2, n):

if (n % i == 0)

break

$\sqrt{n}$ → $\frac{n}{2}$

count = 1
count += 1

$n = 14$ → n
↓
 $14 \div 2 = 7$

316 $\sqrt{n}$

2158 → $\sqrt{n}/2$

PN

13

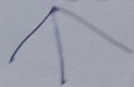
$13 \div 2 = 6.5$

13

13 + 1

13 - 1

* + seconds : 7284



$$7200 + 60 + 24$$

$$\text{then : } 1 \text{ hr} : 24 \text{ sec}$$

$$7284 = n$$

$$\text{hrs} = n // 3600$$

$$\text{min} = (n - \text{hrs} \cdot 3600) // 60$$

$$\text{sec} = (n - \text{hrs} \cdot 3600) \% 60$$

$$7284 // 3600$$

$$= 2$$

$$\text{rem} =$$

$$84$$

$$84 // 60$$

$$= 1$$

$$84 - 1 \cdot 60$$

$$= 24$$

* Check whether the given year is leap year (or) not.

$$\text{year} = 2024$$

$$2024 \cdot 4 = 8096 \neq 0 \text{ not}$$

$$2024 \cdot 4$$

$$\frac{2024}{4} = 506$$

$$2024 \cdot 100$$

$$\frac{2024}{100} \neq 0$$

a year is divisible with 400

or

and not divisible with 100 but divisible with 4.

$$(2, n)$$

$$2 \cdot 1 \cdot 2 = 4$$

$$5 \cdot 1 \cdot 2 = 10$$

* + seconds : 7284

$$7200 + 60 + 24$$

$$\text{time} : 1 \text{ hr} : 24 \text{ sec}$$

$$7284 = n$$

$$\text{hrs} = n // 3600$$

$$\text{min} = (n - 1 \cdot 3600) // 60$$

$$\text{sec} = (n - 1 \cdot 3600) \% 60$$

$$7284 // 3600$$

$$= 2$$

$$\text{rem} = 84$$

$$84 // 60$$

$$= 1$$

$$84 - 1 \cdot 60$$

$$= 24$$

* Check whether the given year is leap year (or) not.

$$\text{year} = 2024$$

$$2024 \% 4 = 0 \text{ not}$$

$$2024 \% 4$$

$$\frac{2024}{4} = 506$$

$$2024 // 100$$

$$\frac{2024}{100} \neq 0$$

a year is divisible with 400

or

and not divisible with 100 but divisible with 4.

$$(2, n)$$

$$2 \cdot 1 \cdot 2 = 0$$

$$5 \cdot 1 \cdot 2 = 0$$